Phase 2 Report

Project Title

PREDICTIVE ANALYTICS FOR URBAN WATER DEMAND

Problem Id: 06

Name: Nitin Singh

University Roll No: 2028786

Sec: Q

Name: Rohan Prasad Tiwari

University Roll No: 2028825

Sec: Q

Name: Ashutosh Singh

University Roll No: 2028675

Sec : Q

1. **Problem Description** 3
2. **Architecture** 4
3. **Models Used** 5
4. **Evaluation metrics used** 6
5. **Preliminary Results and Visualizations** 7
6. **References** 9

1.Problem Description

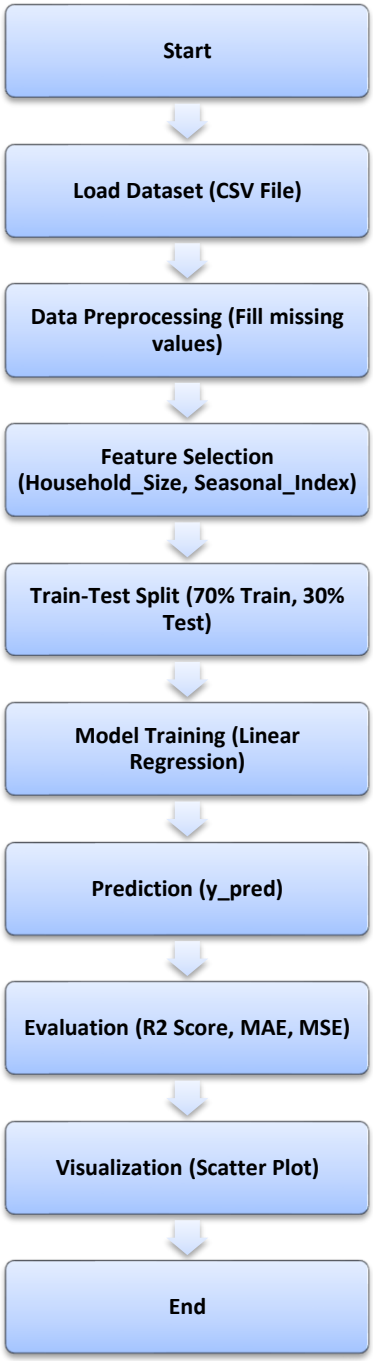
Urban areas are facing increasing challenges in managing water resources due to rapid population growth, changing lifestyles, and seasonal variations. Efficient prediction of daily water consumption is essential for proper planning, conservation, and distribution of water. Traditional estimation methods are often inaccurate and do not consider multiple influencing factors.

This project focuses on developing a machine learning-based system to predict daily water usage using features such as household size, seasonal index, and garden area. By analyzing historical data, the model identifies patterns and relationships between these factors and water consumption.

The system uses data preprocessing, feature selection, and regression techniques to build an accurate prediction model. The results help in understanding water usage trends and improving decision-making.

In real-life applications, this model can assist municipal authorities in optimizing water supply systems, reducing wastage, and ensuring efficient resource management. It can also help households monitor and control their water usage. Overall, this project contributes to sustainable water management and smarter urban planning.

2.Architecture



3.Models Used (Applied/ Planned) Give a brief introduction of the models used

- **Linear Regression Model**

Linear Regression is a supervised machine learning algorithm used to predict continuous values. It establishes a relationship between independent variables (features) and a dependent variable (target) by fitting a linear equation.

In this project, Linear Regression is used to predict daily water consumption based on features such as household size, seasonal index, and garden area. The model learns patterns from the training data and then predicts water usage for new inputs.

The simplicity and efficiency of Linear Regression make it suitable for this project, as it provides good accuracy and is easy to interpret. It also helps in understanding how each feature affects water consumption.

4.Evaluation metrics used

1. **R^2 Score (Coefficient of Determination)**

   R^2 score measures how well the model explains the variance in the target variable. Its value ranges from 0 to 1, where values closer to 1 indicate better model performance.

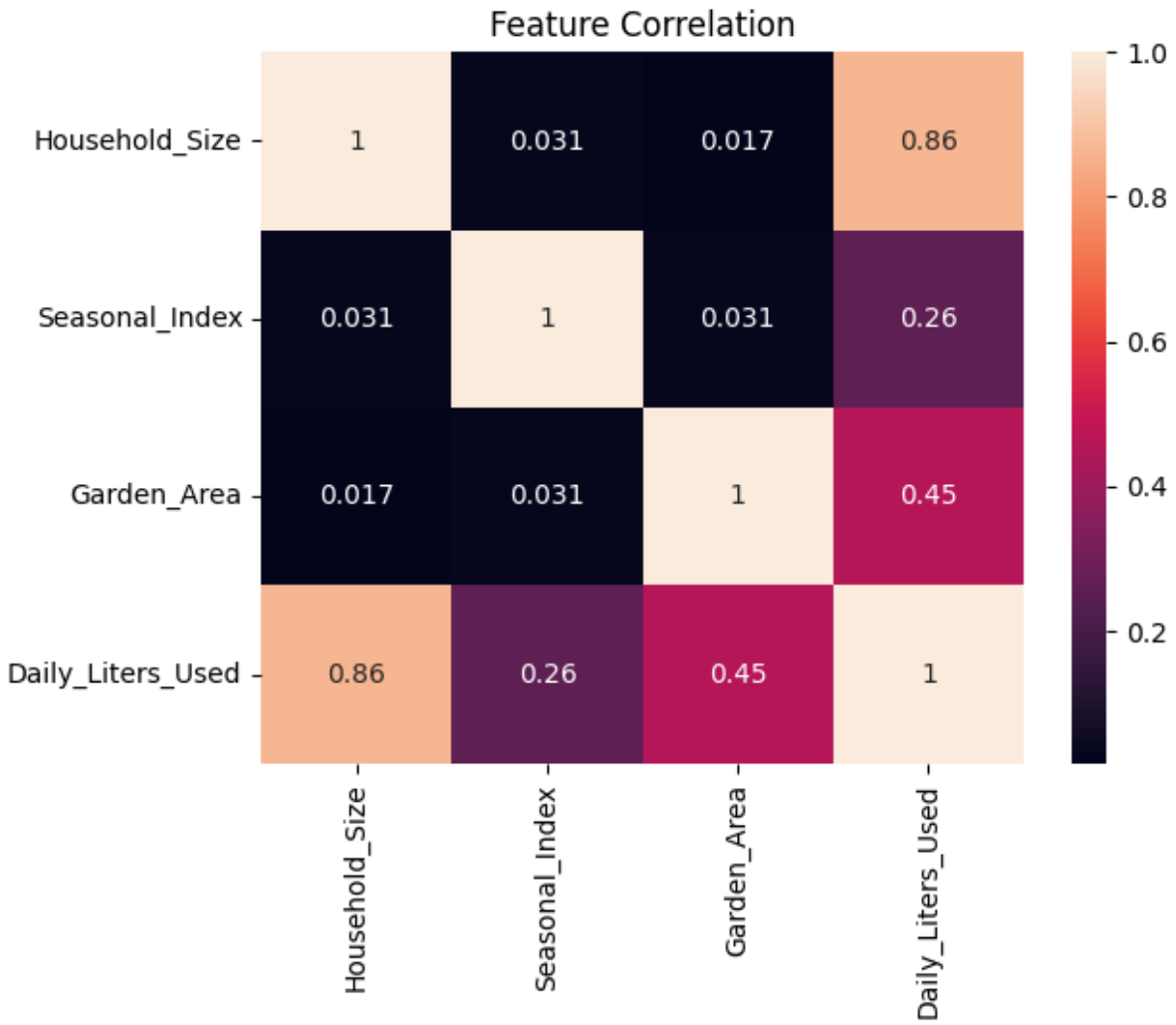
2. **Mean Absolute Error (MAE)**

   MAE calculates the average absolute difference between actual and predicted values. Lower MAE indicates better accuracy.

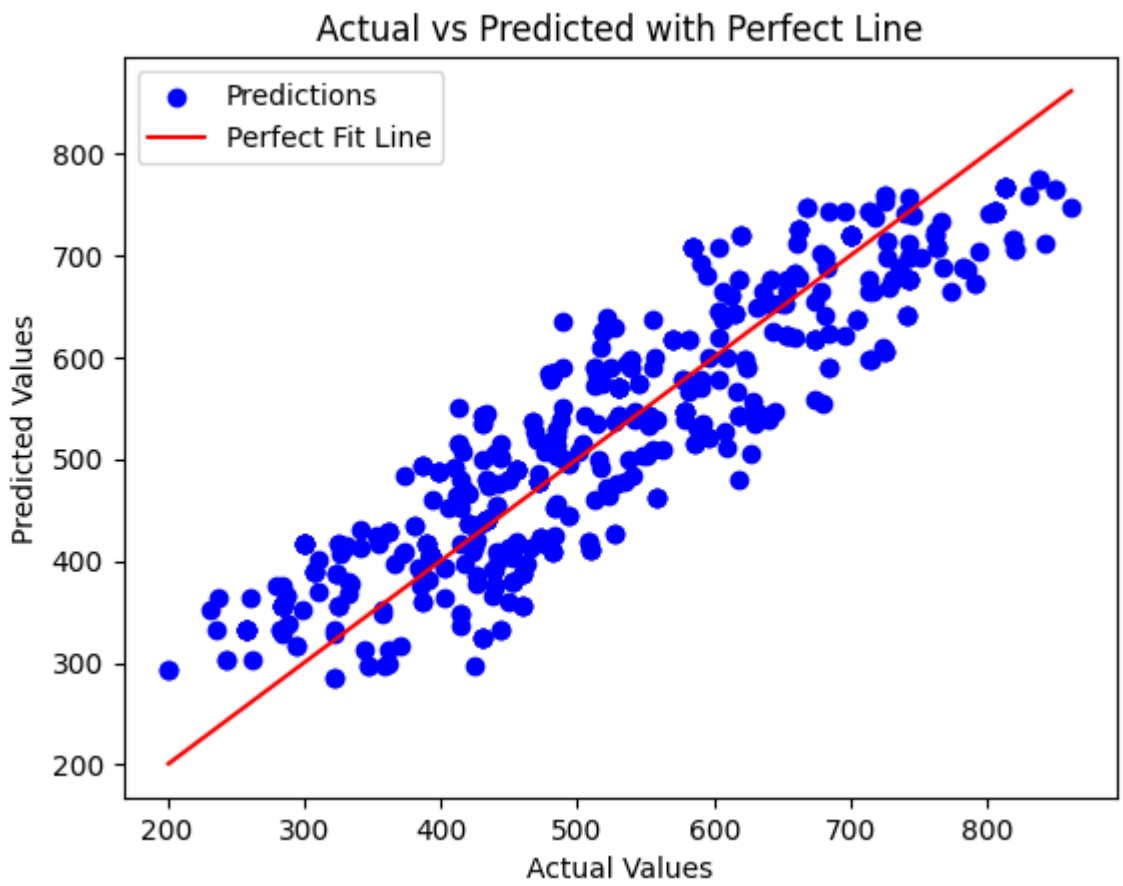
3. **Mean Squared Error (MSE)**

   MSE measures the average squared difference between actual and predicted values. It penalizes larger errors more than MAE. These metrics help in understanding the accuracy and reliability of the prediction model.

5. Preliminary Results and Visualizations



Heatmap showing correlation.



R2 Score: 0.81
MAE: 56.32
MSE: 4320.41
Accuracy: 81.1 %

6. References:

Books:

1. *Let Us Python* – Yashavant Kanetkar
2. *Hands-On Machine Learning with Scikit-Learn* – Aurélien Géron

Online Resources:

1. https://www.geeksforgeeks.org/
2. https://scikit-learn.org/